# Antenna Tool (ANT)
# User guide

**Revision: 2.9**

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# 3. Document history

| Revision | Date | Author | Changes |
|---|---|---|---|
| 0.1 | 02/2017 | Hanit Turjeman | First draft version of the document |
| 0.2 | 03/2017 | Gregory Bugaevsky | Added association by BSSID |
| 0.3 | 05/2017 | Hanit Turjeman | Updated screenshots<br>Added support for 9560/9461 |
| 0.4 | 06/2017 | Hanit Turjeman | Added support for DCR 5166 new Extended HID sensor |
| 0.5 | 07/2017 | Hanit Turjeman | Added antenna isolation feature (DCR 5128) |
| 0.6 | 09/2017 | Hanit Turjeman | Removed antenna C |
| 0.7 | 10/2017 | Hanit Turjeman | Added support for 9462<br>Added "-password <PASS>" API |
| 1.0 | 07/2018 | Gil Volovich | Updated get NVM data feature |
| 1.1 | 10/2018 | Hanit Turjeman | Added support for DCR88 – Scan all Aps on a specific channel and present RSSI measurements. |
| 1.2 | 12/2018 | Gil Volovich | DCR 78- Added support for BIOS SPLC value display |
| 1.3 | 12/2018 | Gil Volovich | DCR 236 - Added support for BT BIOS SAR table display |
| 1.4 | 03/2019 | Nal Zazi | Updated tool to report Chains instead of Antennas |
| 1.5 | 05/2019 | Nal Zazi | Updated supported devices for BT RSSI Test |
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| 1.7 | 09/2019 | Nal Zazi | DCR-359 – Added support for Antenna Gain display |
| 1.8 | 01/2020 | Yael Amran | Updated screenshots |
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| 2.4 | 11/2021 | Liat Abramski | Added support for AX411, windows 11 |
| 2.5 | 03/2022 | Yossi Wortzel | Added support for AX204 |
| 2.6 | 08/2022 | Gleb Nisman | DCR-1595 – Added support for Per country 6E BIOS settings |
| 2.6 | 08/2022 | Yael Amran | Added support for GfP HW manufacture |
| 2.7 | 11/2022 | Yael Amran | Added support for AX203, AX101 and removed AX204<br>Added details for Get Board ID<br>Updated HW manufacture |
| 2.8 | 11/2022 | Ariel Tzabari | Updated user guide for new installers |
| 2.9 | 03/2023 | Yael Amran | Updated TAS info detailes |

# 4. About this publication

## 4.1 Purpose

This document provides information about Intel® ANT application. This document describes ANT setup, installation, and user guidelines.

## 4.2 Intended Audience

This information is intended for Original Equipment Manufacturers (OEMs).

# 5. ANT description

## 5.1 Overview

The Antenna Tool (ANT) is a Command Line Interface (CLI) application used to associate with an access point and optionally perform a RSSI (Received Signal Strength Indicator) test.

ANT performs basic connectivity tests:

- Associating with AP.
- Measuring RSSI.

## 5.2 High level flows

- Association and RSSI test:
  - Connecting to NDIS device.
  - Present RF-Kill status
  - Associating to a specific AP (defined by the user).
  - Measuring RSSI (once RSSI is stable).
  - Displaying test results.
  - Disassociate from AP.
- Single channel scan and RSSI test (unassociated):
  - Disassociate from AP.
  - Connecting to NDIS device.
  - Present RF-Kill status
  - For each iteration (number of scans):
    - Scan all APs found on a specific channel (defined by the user).
    - Get RSSI measurements for each AP on each antenna.
  - Displaying test results.

## 5.3 Setup requirements

### Supported hardware

- Intel® Wireless-AC 8260, 8265, 9260, 9560, 9461, 9462
- Intel® Wi-Fi 6 AX200 160MHz, AX201 160MHz, AX203, AX101 160Mhz
- Intel® Wi-Fi 6E AX210 160MHz, AX211 160MHz, AX411 160MHz

### Operating system requirements

- Windows 10
- Windows 11

# 6. Installation

## 6.1 Prerequisites

Connect supported Intel® Wireless NIC to setup

## 6.2 Installing ANT

Run *ANT_Installer.msi* . Installation wizard will open. Follow the instructions of the installation wizard.

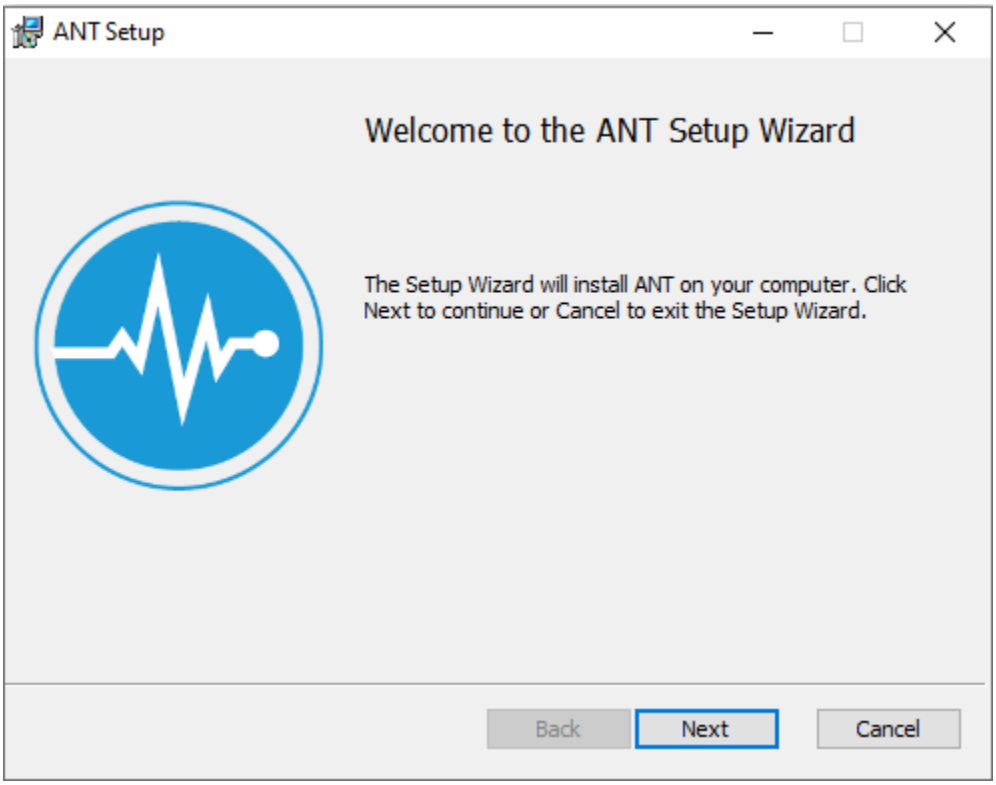



**Figure 1: Installation wizard, welcome screen**

Click “Next” at the first wizard screen.

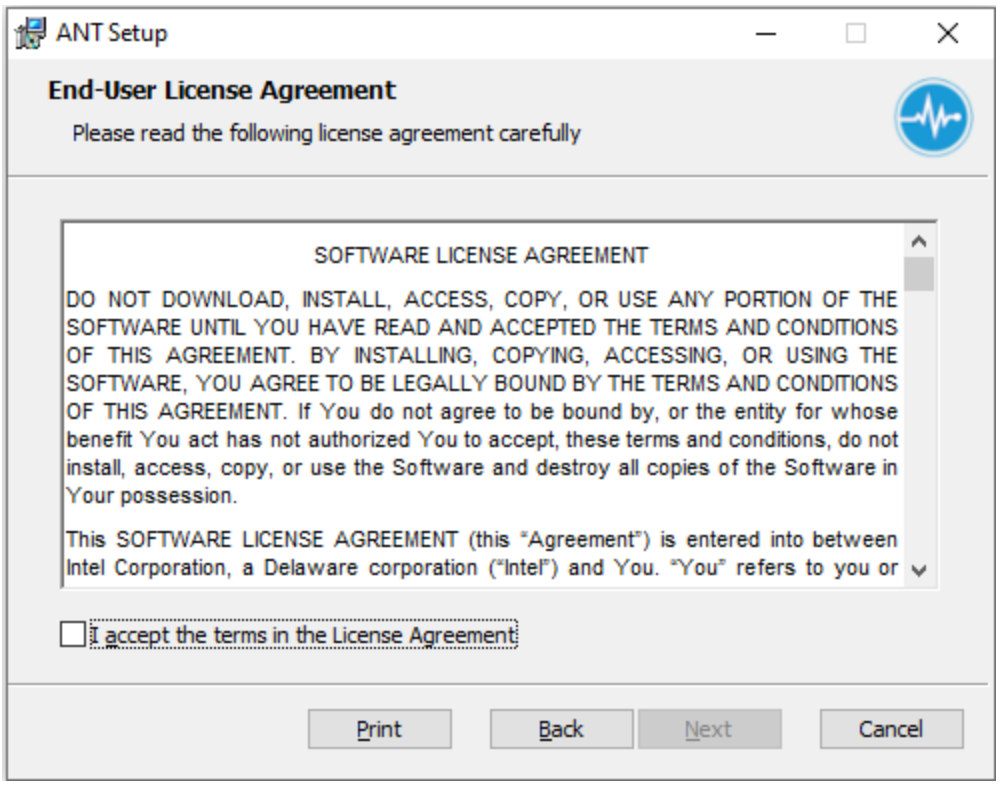



**Figure 2: Installation wizard, Terms and conditions screen**

Accept the terms and conditions.
Click “Next”.

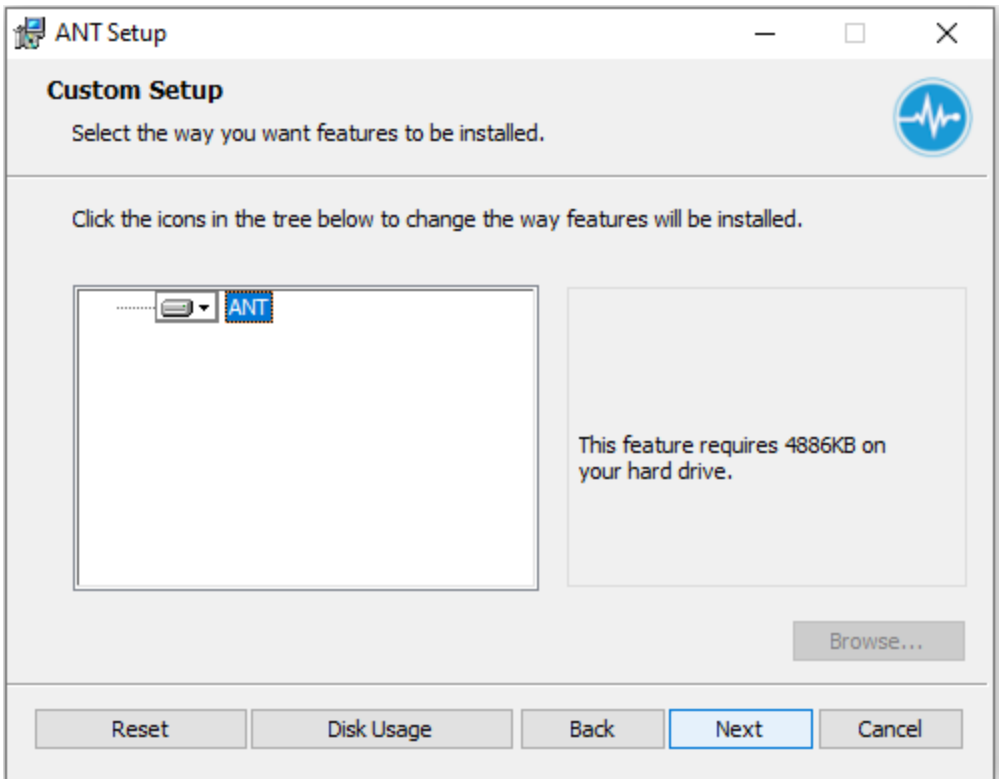



**Figure 3: Installation wizard, Custom setup**

Click “Next”.

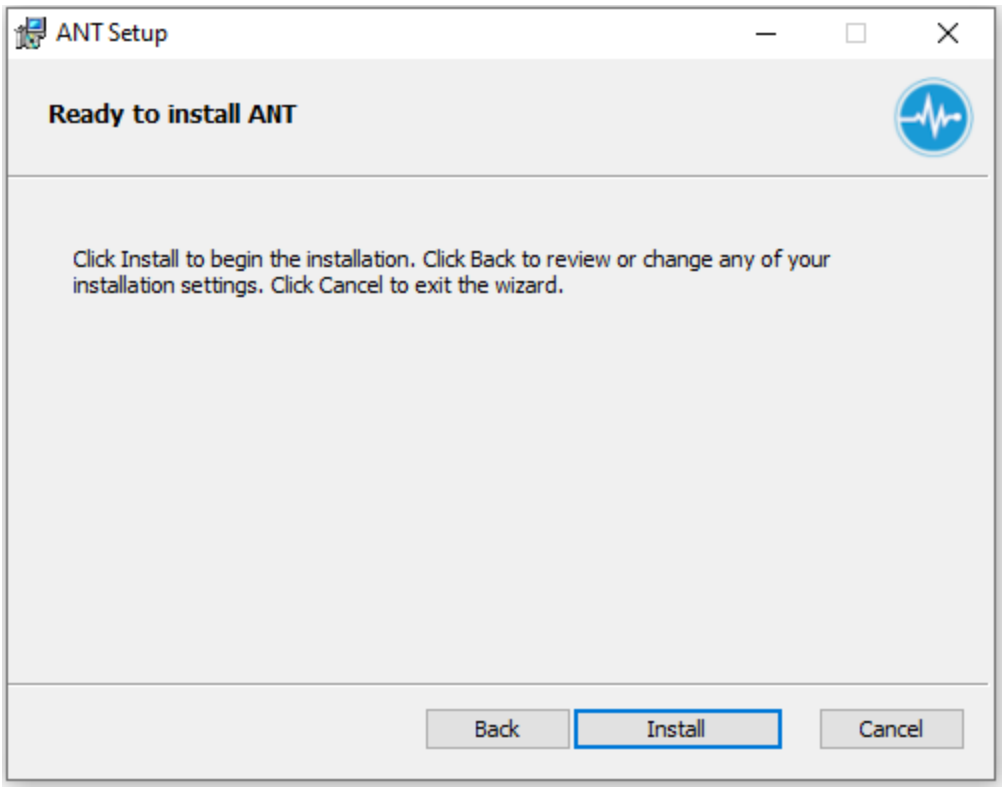
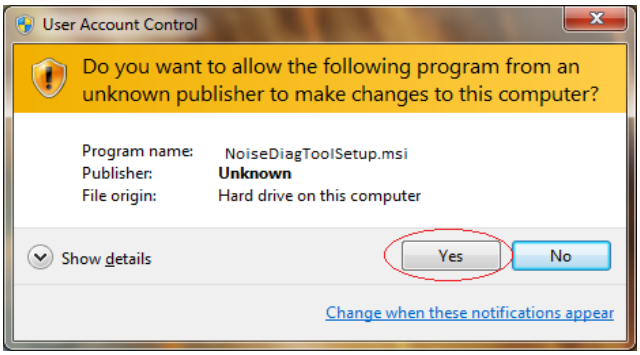
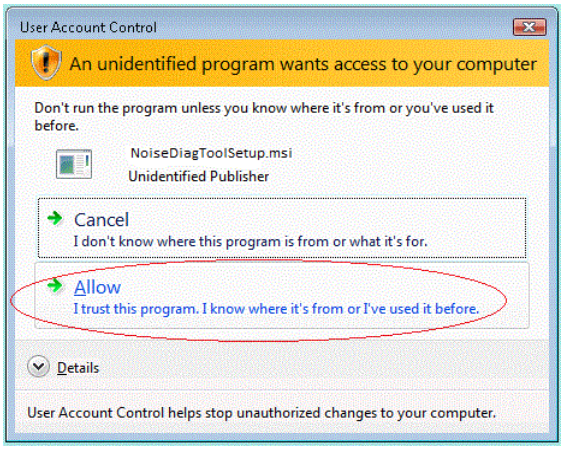



Figure 4: Ready to install

Click “Install” in order to proceed with installation. Click “Back” in order to change installation settings before installation.

**Please note**: during the installation windows might show UAC (User Account Control) pop-up dialog box. In order to continue with the installation please press “**Yes”** or “**Allow”**.



The tool will be installed to "**C:\Program Files\Intel Corporation\ANT**".

**To uninstall the ANT Tool on Microsoft Windows® OS's, take these steps:**

1. Go to **Start**, select **Settings**, and go to **Control Panel** option.
2. Select "**Add or Remove Programs**".
3. Select the "**ANT"** and press the "Remove" button.

## 6.3 Installing ANT from command line

Alternatively you can run the installation from command line. Make sure to open cmd console with administrative privileges. Run the following command:
"ANT_Installer.msi /quiet /qn /norestart"

**In order to complete the installation correctly, restarting the computer is required before executing ANT.**

# 7. Running

## 7.1 Prerequisites

- Supported Intel® Wireless NIC is connected to the setup.
- Intel® Wifi driver is installed and NIC is operational (enabled). ANT release doesn't include driver package.

## 7.2 Limitations

- ANT does not accept any unsupported command line parameters.
- Command line commands/parameters are not case sensitive except for SSID name (-network <ssid>).
- ANT should run using administrative privilege.
- The SSID name should be surrounded by double quote marks if it contains space symbol.
- You have to disable/enable wireless NIC after all tests done because ANT changes default driver settings.
- An antenna option (one of the following: -ant_all, -antd, -ant1, -ant2) must be passed in order to enable RSSI testing.
- "-band" option changes the default band settings of the driver. By default, driver works in mixed mode (listens to all bands). If you don't need to change this behavior – don't apply "band" flag. In order to restore mixed band listening, you should run "ant –reset_driver".
- "Chains" reports list of chains, marked by driver as "active", after the first succeeded association. Notice, that driver will run detection algorithm only once, till next driver reset (disable/enable or fresh installation of a driver). Therefore, any chain manipulation after the first association and till next driver reset will not affect driver settings, and will not be reflected by ANT tool as well.

## 7.3 Steps to run ANT

- Power up the access point.
- Copy ANT layout to a local folder.
- Run ant.exe from a CMD window with the desired command line options:
    - Type "ant -?" for usage if you are not familiar with the tool.
    - Type "ant <command line parameters ...>" to run the tool.

❖ Running the tool with no parameters will display a help screen.

# 8. ANT functionalities

## 8.1 Association

Performing association with an access point (by SSID and optionally by BSSID).

Command line options:
ANT –network <ssid> [-bssid <bssid>] [-password <pass>] [-band <a|b|g>] [-enable_11n | -disable_11n] [-stay_assoc]
[-ant_isolation] [-delay <value>] [-aggressive] [-d | -debug <append | overwrite>] [–log <filename>]

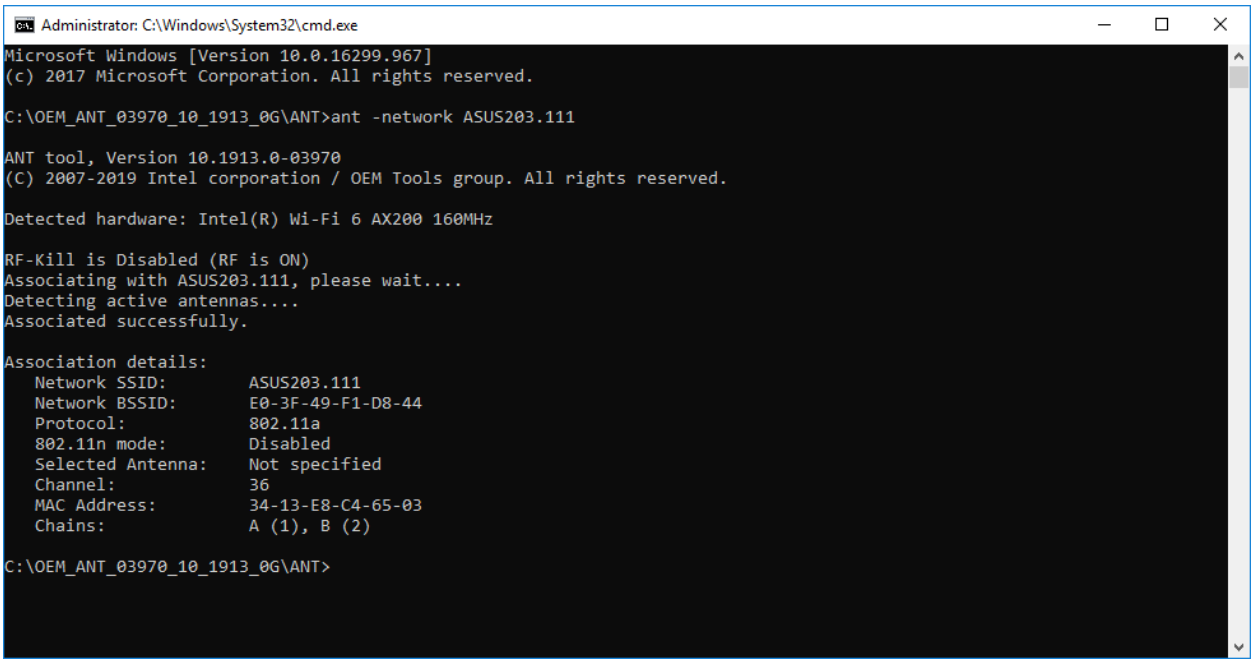



Figure 5: Association

```
Administrator: C:\Windows\System32\cmd.exe

C:\OEM_ANT_03970_10_1913_0G\ANT>ant -network ASUS203.111 -bssid e0-3f-49-f1-d8-40

ANT tool, Version 10.1913.0-03970
(C) 2007-2019 Intel corporation / OEM Tools group. All rights reserved.

Detected hardware: Intel(R) Wi-Fi 6 AX200 160MHz

RF-Kill is Disabled (RF is ON)
Associating with e0-3f-49-f1-d8-40, please wait....
Detecting active antennas....
Associated successfully.

Association details:
   Network SSID:        ASUS203.111
   Network BSSID:       E0-3F-49-F1-D8-40
   Protocol:            802.11g
   802.11n mode:        Disabled
   Selected Antenna:    Not specified
   Channel:             6
   MAC Address:         34-13-E8-C4-65-03
   Chains:              A (1), B (2)

C:\OEM_ANT_03970_10_1913_0G\ANT>
```

**Figure 6: Association with BSSID**

## 8.2 RSSI test

Performing association with an access point (by SSID and optionally by BSSID) and executes RSSI test.

Command line options:
ANT –network <ssid> [-bssid <bssid>] [-password <pass>] –antd|-ant1|-ant2 |-ant_all [-band <a|b|g>]
[-I <iterations>] [-th <threshold>] [-th_high <threshold>] [-stay_assoc] [-enable_11n | -disable_11n]
[-rssi_loops <loops count>] [-falloff <value>] [-stable_rssi_threshold <value>] [-ant_isolation]
[-delay <value>] [-aggressive] [-xml <a|o>] [-OutputXMLFile <filename>]
[-d | -debug <append | overwrite>] [–log <filename>]

- When the test is done, the NIC will be disconnected. Use [-stay_assoc] to stay associated.

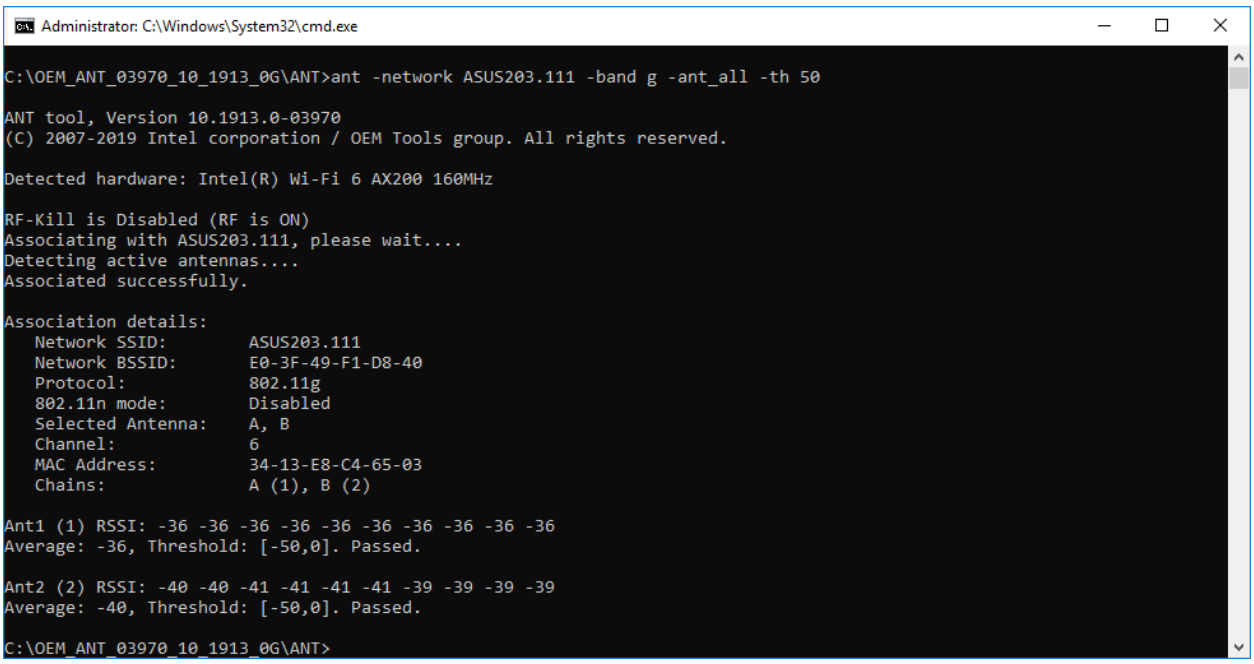
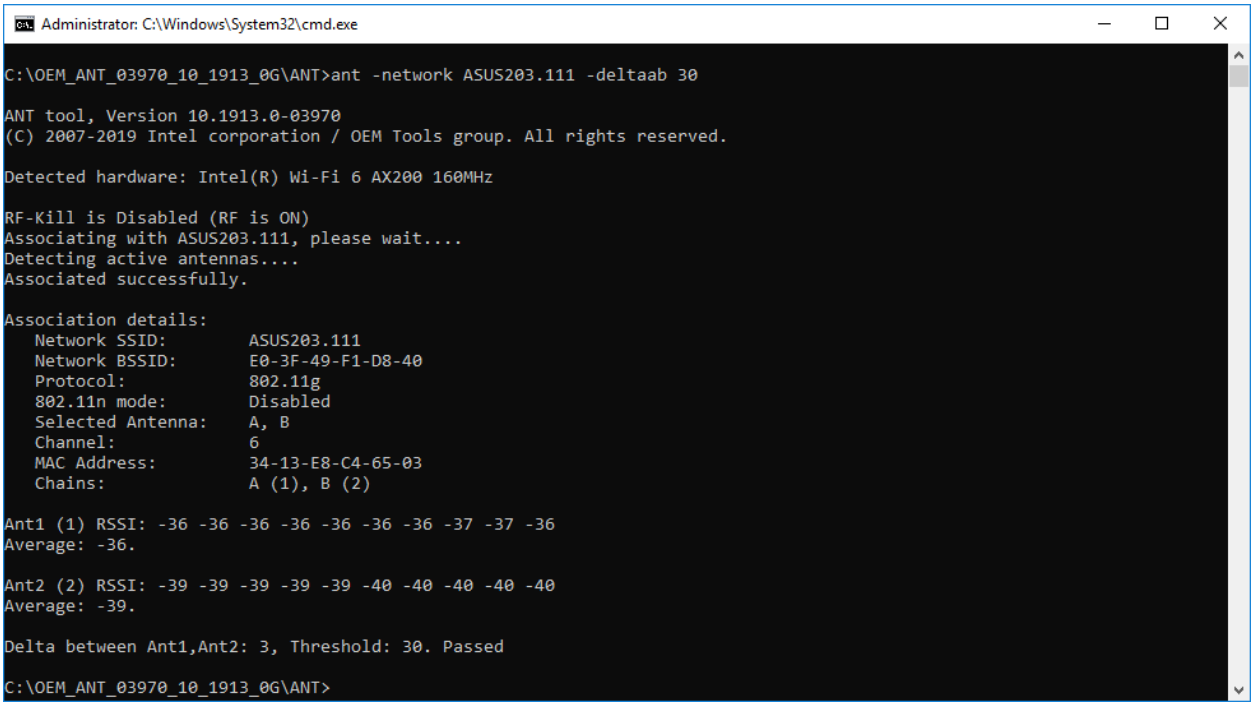



Figure 7: RSSI test

## 8.3 Delta test

Performs association with an access point and executes RSSI delta test between two chains.

Command line options:
ANT –network <ssid> [-bssid <bssid>] [-password <pass>] –deltaab|-deltaac|-deltabc <threshold> [-band <a|b|g>] [-stay_assoc] [-enable_11n | -disable_11n] [-rssi_loops <loops count>] [-falloff <value>] [-stable_rssi_threshold <value>] [-ant_isolation] [-delay <value>] [-aggressive] [-xml <a|o>] [-OutputXMLFile <filename>] [-d | -debug <append | overwrite>] [–log <filename>]



```
Administrator: C:\Windows\System32\cmd.exe

C:\OEM_ANT_03970_10_1913_0G\ANT>ant -network ASUS203.111 -deltaab 30

ANT tool, Version 10.1913.0-03970
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Detected hardware: Intel(R) Wi-Fi 6 AX200 160MHz

RF-Kill is Disabled (RF is ON)
Associating with ASUS203.111, please wait....
Detecting active antennas....
Associated successfully.

Association details:
   Network SSID:        ASUS203.111
   Network BSSID:       E0-3F-49-F1-D8-40
   Protocol:            802.11g
   802.11n mode:        Disabled
   Selected Antenna:    A, B
   Channel:             6
   MAC Address:         34-13-E8-C4-65-03
   Chains:              A (1), B (2)

Ant1 (1) RSSI: -36 -36 -36 -36 -36 -36 -36 -37 -37 -36
Average: -36.

Ant2 (2) RSSI: -39 -39 -39 -39 -39 -40 -40 -40 -40 -40
Average: -39.

Delta between Ant1,Ant2: 3, Threshold: 30. Passed

C:\OEM_ANT_03970_10_1913_0G\ANT>
```

Figure 8: Delta test

## 8.4 Single channel scan and RSSI test

In single channel scan mode ANT is requested to survey defined channel and notify the user on APs found and their RSSI value per antenna. The user must provide the desired channel. The scan is active only, thus only active channels are supported.
Acceptable channels: 1-11, 36, 40, 44, 48, 149, 153, 157, 161 and 165
In addition, the user can specify:

- Iteration number of RSSI measurement (iteration number means number of scans):
    - Allowed values: 1 – 100. Default is 10.
- Time interval of every iteration in millisecond (ΔT –time between scans):
    - Allowed values: 200,400,600,800 and 1000. Default is 200.

While using scan mode, user shall consider the option that some APs will not be presented on the results table for all of the iterations. To find APs, ANT will trigger a probe request transmission and wait time of ~20ms on each iteration. ANT will present All APs in which it could capture their probe response during this 20 millisecond time, any beacon that would be received will also be added to the found APs list. Due to this behavior, APs which cannot Rx the Intel Device transmission, but are received by the Intel device, will only be represented on results table on few cases.

Scan result values for a specific iteration can be one of the following:

- RSSI value [-95…0]
- N/A – AP was not found.
- ERR – Scan was failed.

Command line options:
ANT –scan –channel <active channel number> [-rssi_loops <loops count>] –ant1|-ant2 |-ant_all
[-interval <time in ms>] [-th <threshold>] [-th_high <threshold>]
[-d | -debug <append | overwrite>] [–log <filename>]

```
Administrator: C:\Windows\System32\cmd.exe

C:\OEM_ANT_03970_10_1913_0G\ANT>ant -scan -channel 6

ANT tool, Version 10.1913.0-03970
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Detected hardware: Intel(R) Wi-Fi 6 AX200 160MHz

RF-Kill is Disabled (RF is ON)
Detecting active antennas....
ANT scan result.

   Selected Antenna:    A, B
   Channel:             6
   Interval(ms):        200
   Iterations:          10
   Chains:              A (1), B (2)
--------------------------------------------------
SSID 1:          ASUS203.111
BSSID 1:         E0-3F-49-F1-D8-40


Ant1 (1) RSSI: -38 -37 -36 -37 -36 -36 -33 -37 -37 -36
Average: -36, Threshold: [-45,0]. Passed.

Ant2 (2) RSSI: -39 -40 -40 -40 -40 -39 -37 -40 -39 -39
Average: -39, Threshold: [-45,0]. Passed.
--------------------------------------------------
SSID 2:          ASUS_2.4G
BSSID 2:         2C-4D-54-74-CF-30
```
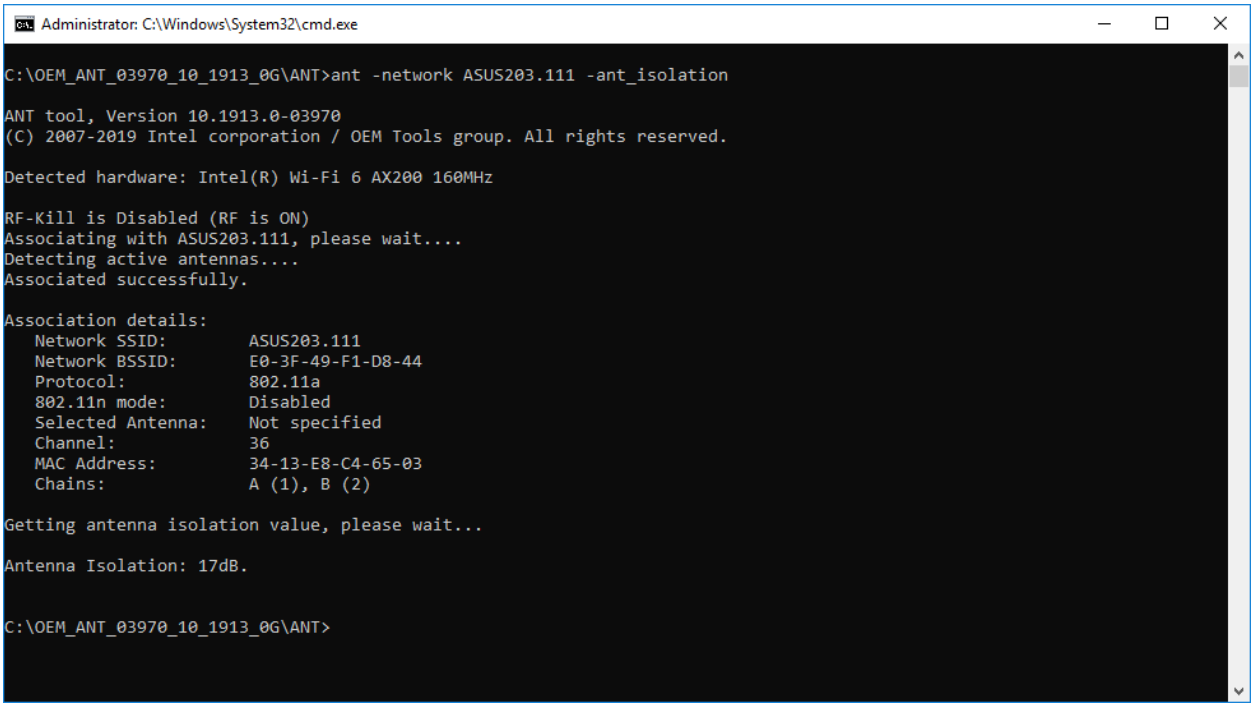

Figure 9: Single channel scan and RSSI test

## 8.5 Antenna isolation

Antenna isolation value represent the physical isolation between the two antennas. This value is calculated periodically by the Wi-Fi FW when Wi-Fi is associated to AP (**2.4 GHz only**) and BT is active.
Antenna isolation value is in dB.
Antenna isolation value is highly important in the context of Wi-Fi-BT Coex, when each core use a separate antenna.
In order to get antenna isolation value user should add the parameter "-ant_isolation" to association, RSSI or delta test. For examples: ANT –network <SSID> -ant_isolation



```
Administrator: C:\Windows\System32\cmd.exe

C:\OEM_ANT_03970_10_1913_0G\ANT>ant -network ASUS203.111 -ant_isolation

ANT tool, Version 10.1913.0-03970
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Detected hardware: Intel(R) Wi-Fi 6 AX200 160MHz

RF-Kill is Disabled (RF is ON)
Associating with ASUS203.111, please wait....
Detecting active antennas....
Associated successfully.

Association details:
   Network SSID:        ASUS203.111
   Network BSSID:       E0-3F-49-F1-D8-44
   Protocol:            802.11a
   802.11n mode:        Disabled
   Selected Antenna:    Not specified
   Channel:             36
   MAC Address:         34-13-E8-C4-65-03
   Chains:              A (1), B (2)

Getting antenna isolation value, please wait...

Antenna Isolation: 17dB.


C:\OEM_ANT_03970_10_1913_0G\ANT>
```

**Figure 10: Antenna isolation value**

## 8.6 Driver control

Allows to enable, disable or reset NIC's driver.

Command line options:

- Wi-Fi Driver: ANT –enable_driver|-disable_driver|-reset_driver
- Bluetooth Driver: ANT –enable_bt_driver|-disable_bt_driver

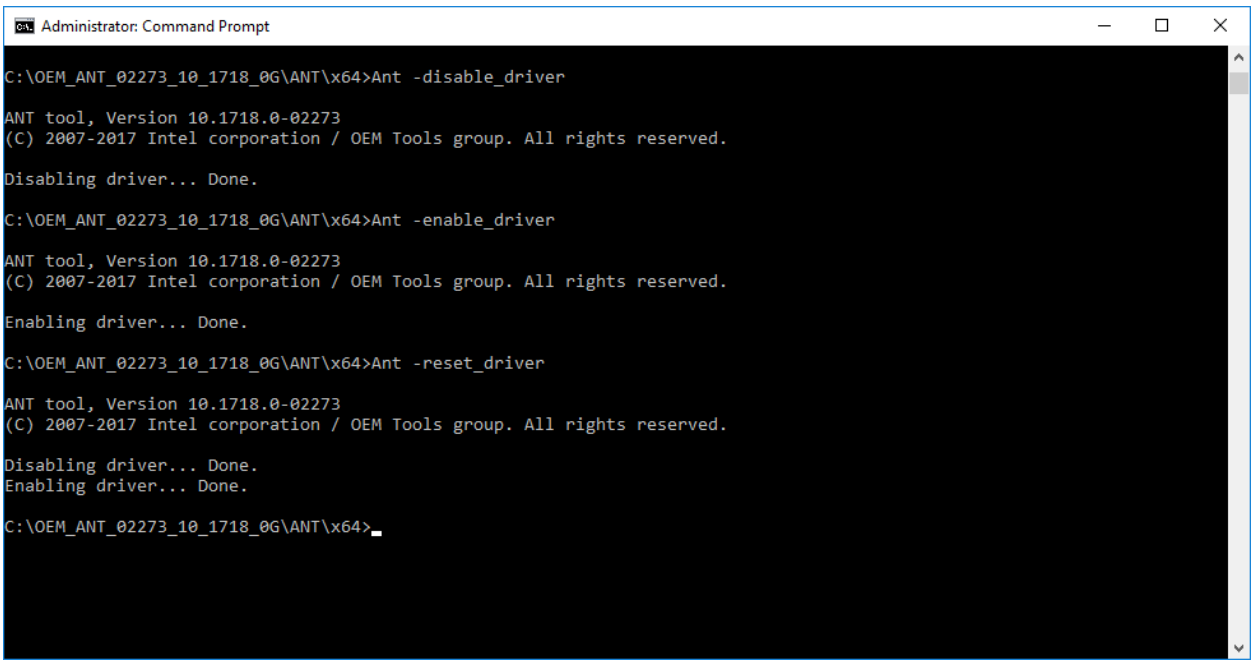



Figure 11: Driver control, Disable, Enable and Reset Wi-Fi driver

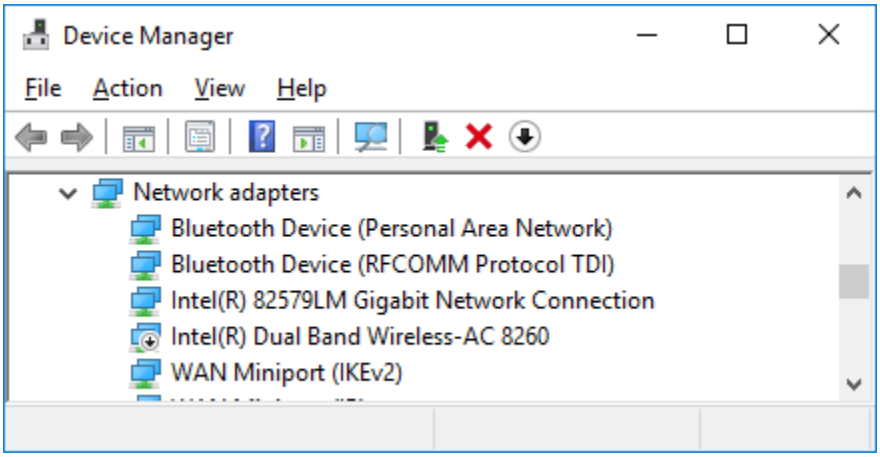



Figure 12: Driver control, Wi-Fi driver disabled

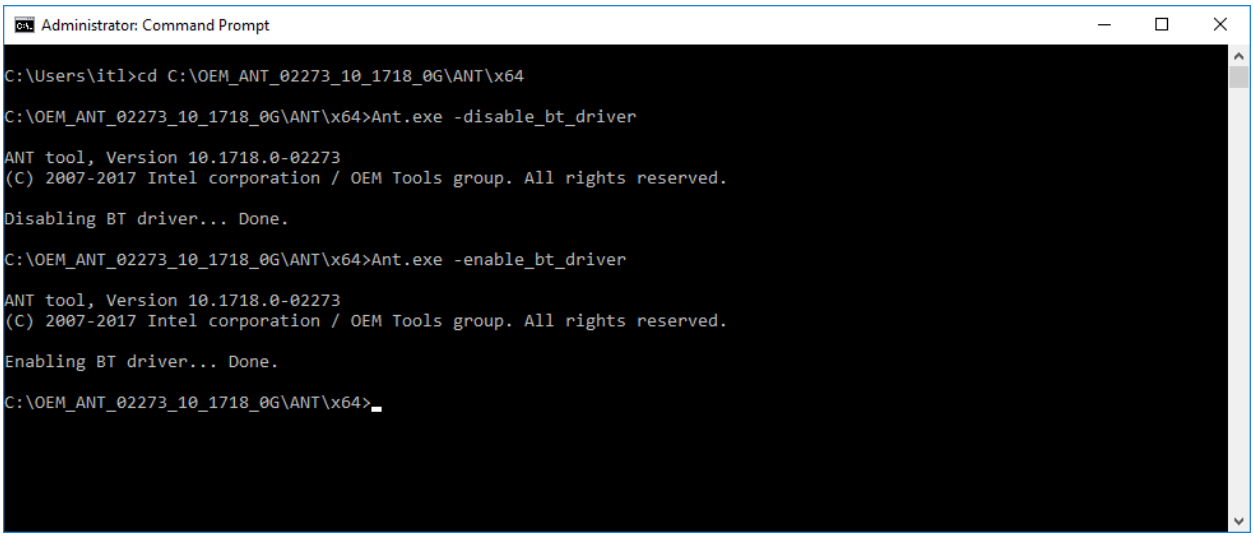



Figure 13: Driver control, Disable, Enable BT driver

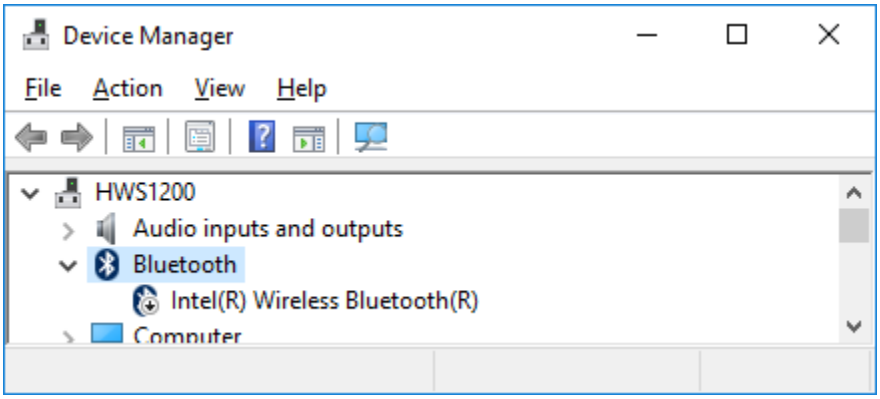



Figure 14: Driver control, BT driver disabled

## 8.7 HW RF Kill status

Returns the RF Kill switch status.

Command line options: ANT –get_rf_kill

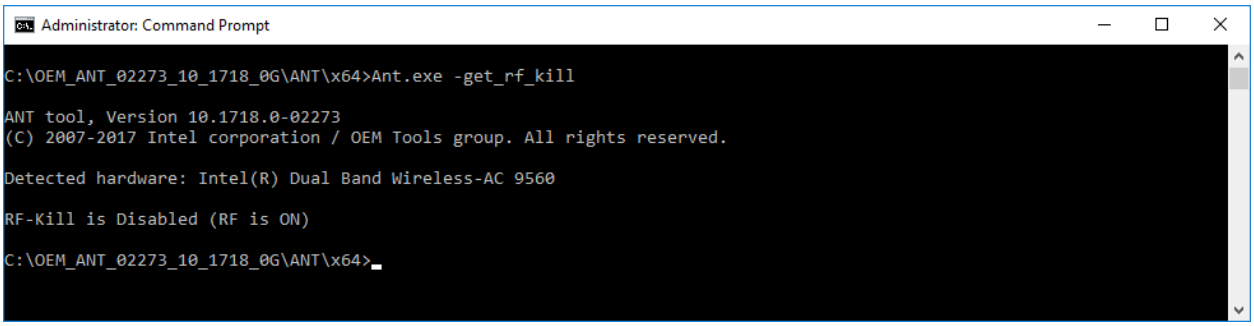



Figure 15: RF-Kill status

## 8.8 Check HW version

Shows / compares hardware version. Supported only for WP device.

Command line options:
ANT –get_hw_ver|-compare_hw_ver <HW_VERSION_COMPARING_WITH>

## 8.9 Check device ID

Shows device ID.

Command line options:
ANT –get_deviceid

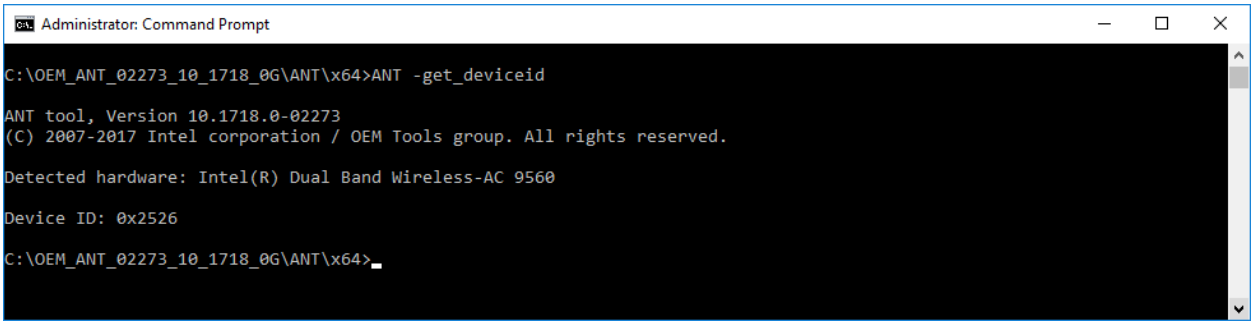



Figure 16: Device ID

## 8.10 Check subsystem ID

Shows subsystem ID.

Command line options:
ANT –get_subsystemid

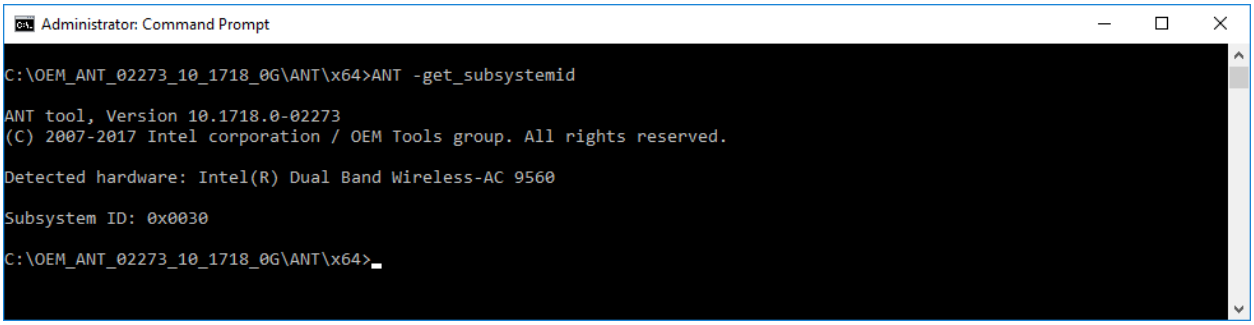



Figure 17: Subsystem ID

## 8.11 Check NVM image version

Shows NVM image version.

Command line options:
ANT –get_nvmimage

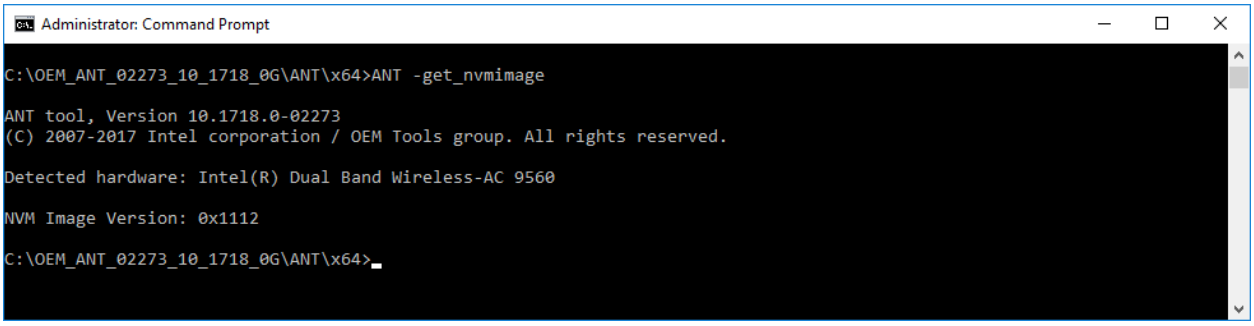



Figure 18: NVM image version

## 8.12 Check OTP OEM version

Shows OTPs OEM version.

Command line options:
ANT –get_oemversion

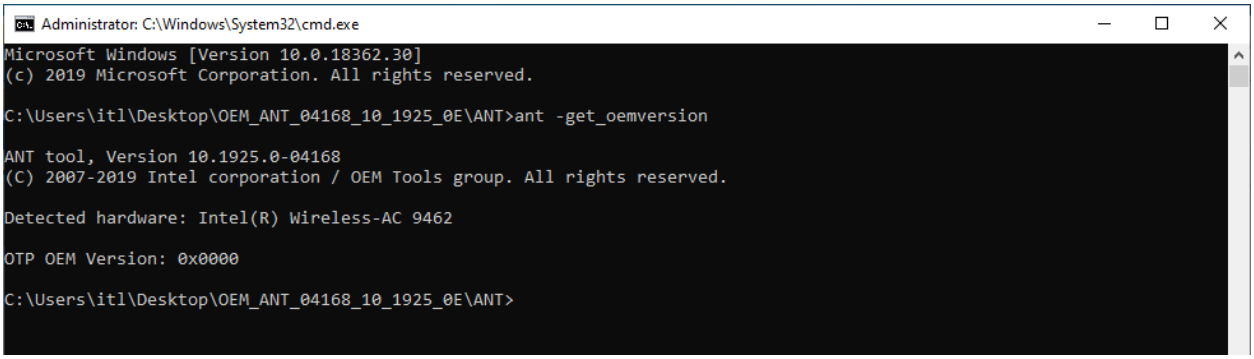



Figure 19: OTP OEM version

## 8.13 Check PCI revision ID

Shows HW revision ID.

Command line options:
ANT –get_pcirevid

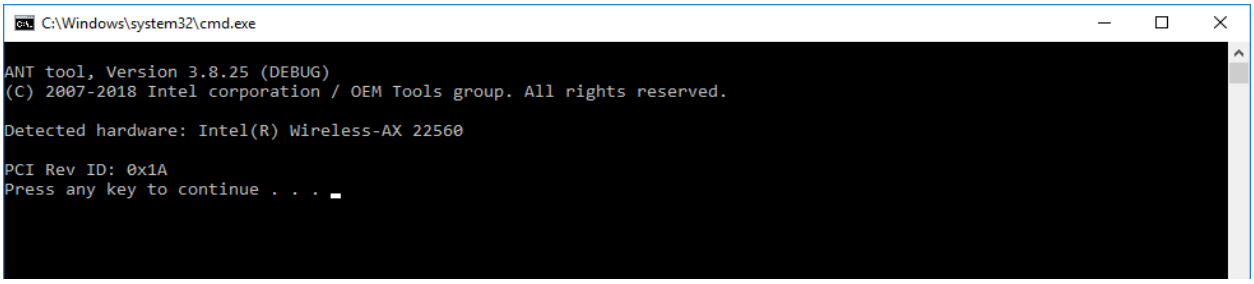



Figure 20: PCI revision ID

## 8.14 Check NVM data

Shows NVM data (all data described in 7.6 – 7.10 sections above).

Command line options:
ANT –get_device_data

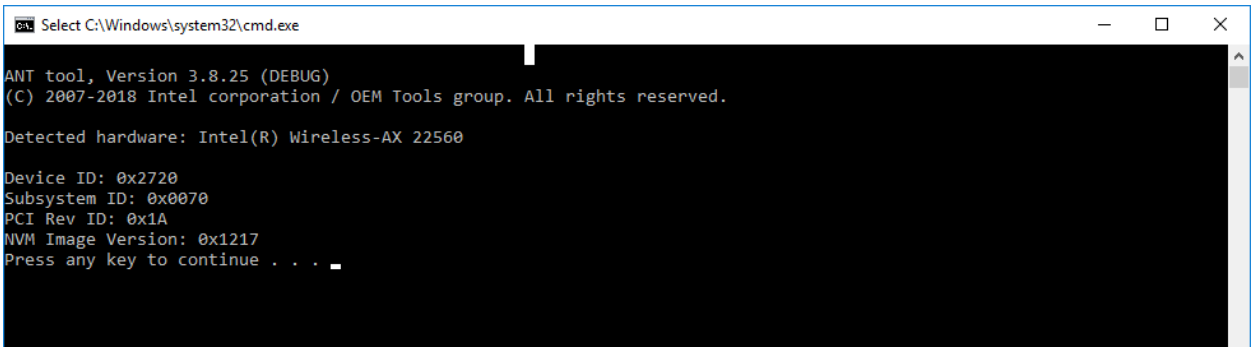



Figure 21: Get device data

## 8.15 Bluetooth RSSI test

Measuring RSSI via BT antenna (Listening ONLY to channel 0). Supported only for WP1, SdP and JfP1 devices.

Command line options:
ANT –BT

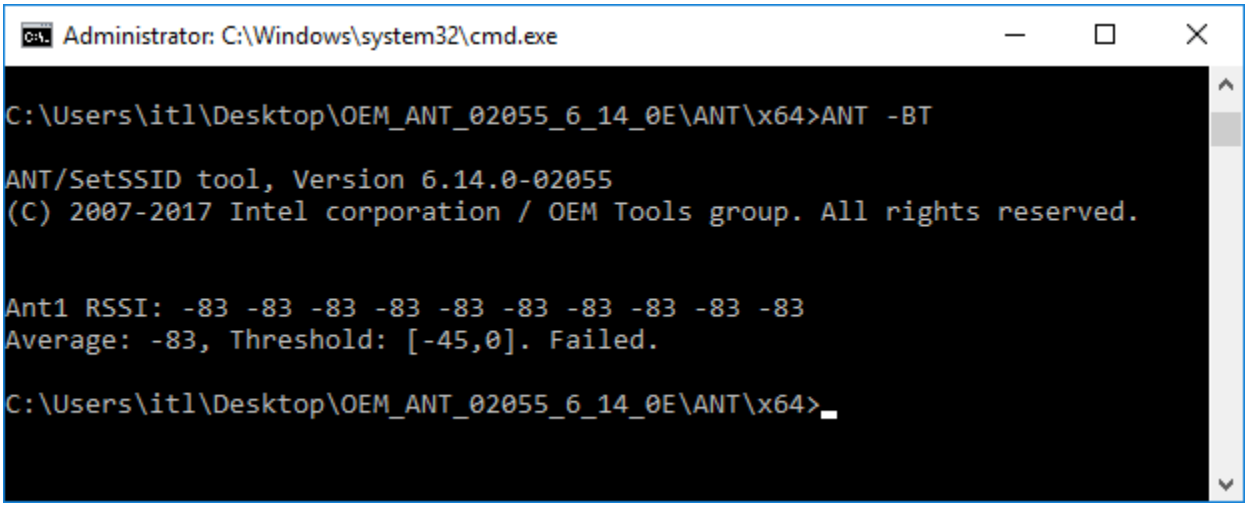



Figure 22: Test RSSI via Bluetooth antenna

## 8.16 Set MCC (Mobile Country Code)

Sets the requested MCC for 2 hours. If the MCC was changed during the 2 hours period, the remaining time will be decided according to first mcc request.
When the 2 hours period will expire the FW will be down for 15 minutes.

Command line options:
ANT –SetMcc <mcc>

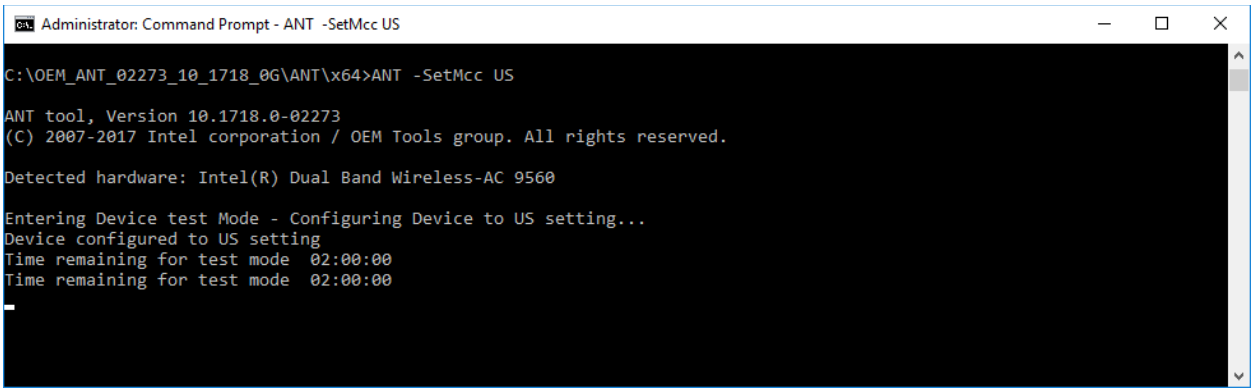



Figure 23: Set MCC

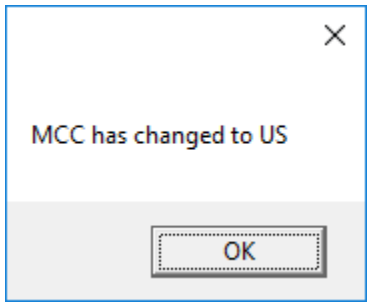



Figure 20: Set MCC, confirm message.

## 8.17 Query MCC

The query will return the following:

1. Current MCC
2. MCC source (OTP, DRS, MCC API…)

   If the source is from MCC API the additional information will be displayed:

   a) Device mode (active, not active, denial).
   b) Remaining time in the current mode (if in active or denial).

   An error level is returned (32 bits) with the following data:

   a) bits 0-15 – Current mcc (if not in active mode will be 0).
   b) bits 16-23 – Device mode (0 – active, 1 – not active, 2 – denial).
   c) bits 24-31 – elapsed time in current mode (in 30 seconds interval. Example: 1 – 30 seconds, 2 - 60 seconds ...)

Command line options:

ANT –QueryMCC

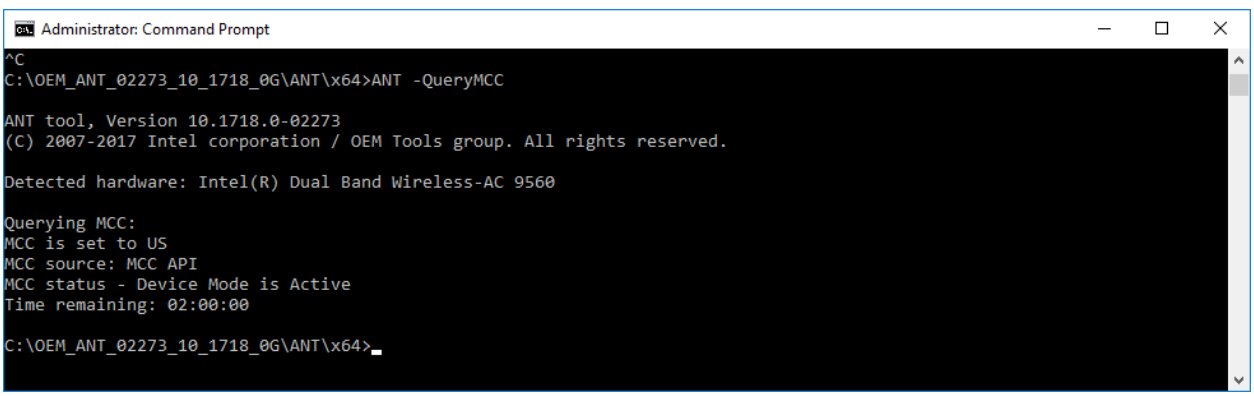



Figure 24: Query MCC

## 8.18 Get Lenovo pSensor Tx Limits

Showing the following information:

- Sensor type (Generic or Lenovo).
- Sensor sub type (WMI, HID, EXTENDED HID…)
- For “WMI” or “HID” sensor:
  - 2.4GHz Tx Power Limit
  - 5.2GHz Tx Power Limit
- For “EXTENDED HID” sensor:
  - For Chain A\B:
    - 2.4GHz
    - 5.2GHz 5150MHz-5350MHz
    - 5.2GHz 5350MHz-5470MHz
    - 5.2GHz 5470MHz-5725MHz
    - 5.2GHz 5725MHz-5950MHz

Command line options:
ANT –get_psensor_tx_limits

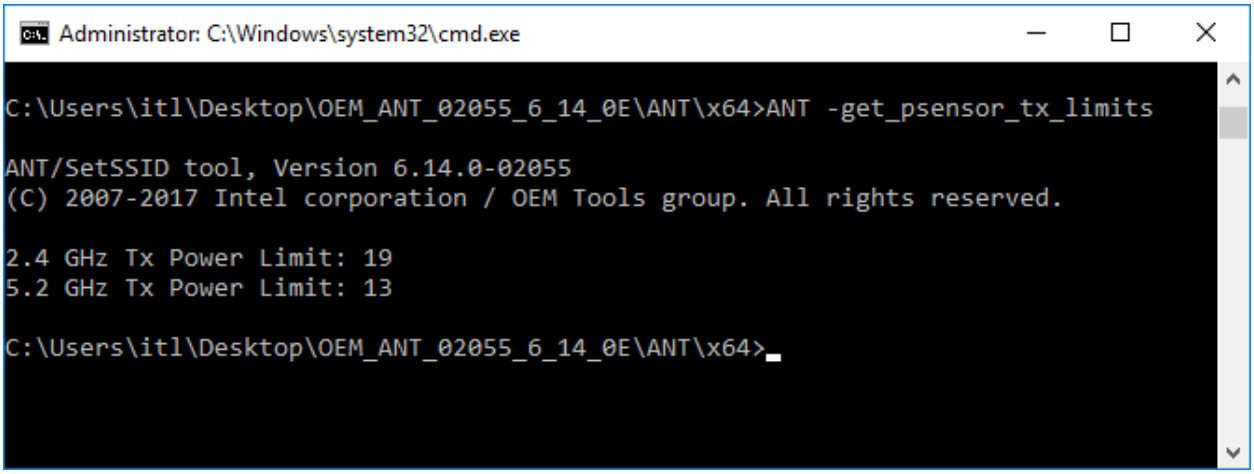



Figure 25: pSensor Tx limits “WMI” or “HID” sensor

```
C:\Windows\system32\cmd.exe

C:\Users\yy\Desktop\gitgat\AntV2>ANT -get_psensor_tx_limits

ANT tool, Version 10.1723.0-02345
(C) 2007-2017 Intel corporation / OEM Tools group. All rights reserved.

Detected hardware: Intel(R) Dual Band Wireless-AC 8265

Sensor type: LENOVO
Sensor sub-type: EXTENDED HID

SAR BIOS Tx power limits:

 Chain A:
        2.4GHz:                 19 dBm (0x98)
        5.2GHz 5150MHz-5350MHz: 14.5 dBm (0x74)
        5.2GHz 5350MHz-5470MHz: 15 dBm (0x78)
        5.2GHz 5470MHz-5725MHz: 14.5 dBm (0x74)
        5.2GHz 5725MHz-5950MHz: 15 dBm (0x78)

 Chain B:
        2.4GHz:                 19 dBm (0x98)
        5.2GHz 5150MHz-5350MHz: 14.5 dBm (0x74)
        5.2GHz 5350MHz-5470MHz: 15 dBm (0x78)
        5.2GHz 5470MHz-5725MHz: 14.5 dBm (0x74)
        5.2GHz 5725MHz-5950MHz: 15 dBm (0x78)


C:\Users\yy\Desktop\gitgat\AntV2>
```

**Figure 26: pSensor Tx limits "EXTENDED HID" sensor**

## 8.19 Get BIOS SAR Table

Showing the following information for Chain A\B:

- 2.4GHz
- 5.2GHz 5150MHz-5350MHz
- 5.2GHz 5350MHz-5470MHz
- 5.2GHz 5470MHz-5725MHz
- 5.2GHz 5725MHz-5950MHz

Command line options:
ANT –get_bios_sar_table

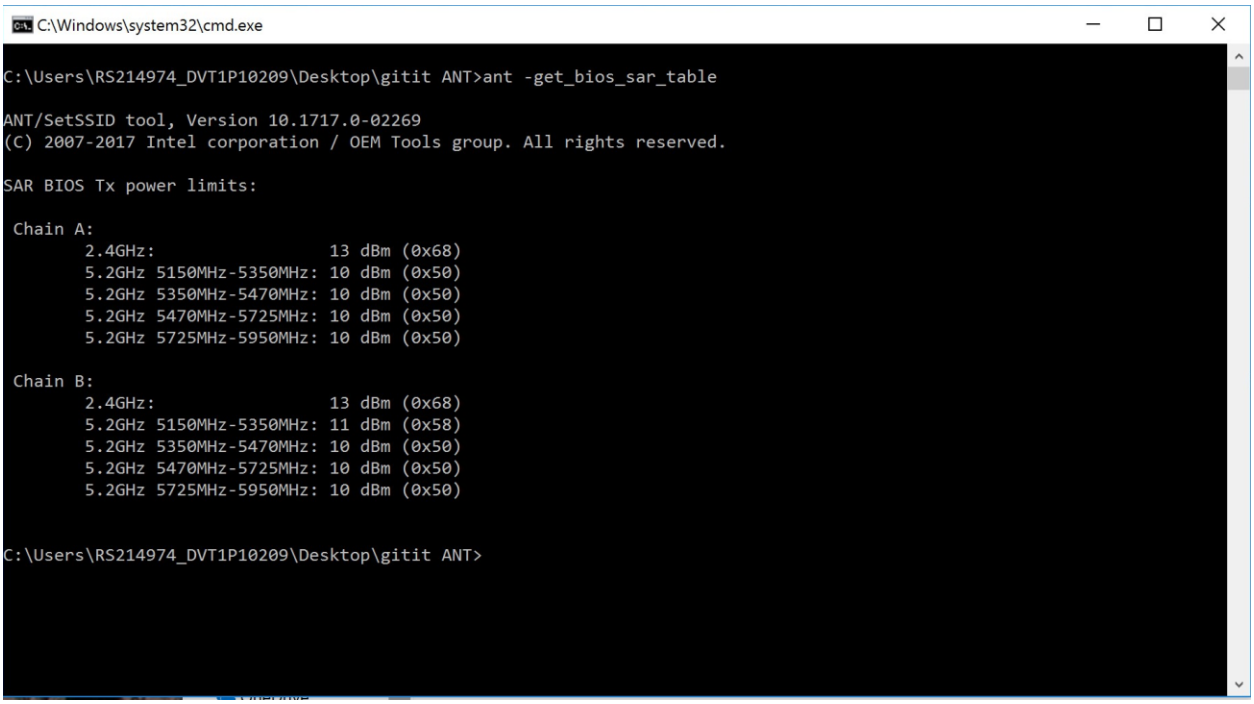



Figure 27: BIOS SAR table

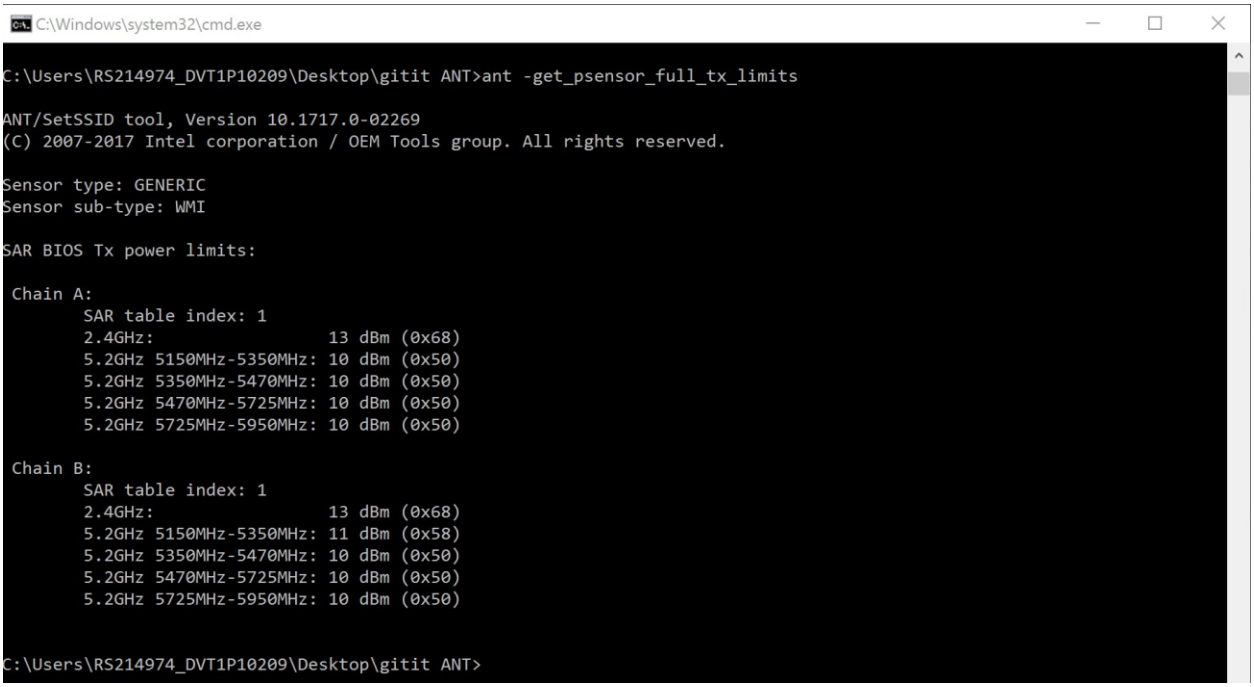
## 8.20 Get Generic pSensor Tx Limits

Showing the following information:

- Sensor type (Generic or Lenovo).
- Sensor sub type (WMI, HID, EXTENDED HID…)
- Chain A/B SAR table in use.
- For Chain A\B:
  - 2.4GHz
  - 5.2GHz 5150MHz-5350MHz
  - 5.2GHz 5350MHz-5470MHz
  - 5.2GHz 5470MHz-5725MHz
  - 5.2GHz 5725MHz-5950MHz

Command line options:

ANT –get_psensor_full_tx_limits



```
C:\Windows\system32\cmd.exe

C:\Users\RS214974_DVT1P10209\Desktop\gitit ANT>ant -get_psensor_full_tx_limits

ANT/SetSSID tool, Version 10.1717.0-02269
(C) 2007-2017 Intel corporation / OEM Tools group. All rights reserved.

Sensor type: GENERIC
Sensor sub-type: WMI

SAR BIOS Tx power limits:

 Chain A:
        SAR table index: 1
        2.4GHz:                 13 dBm (0x68)
        5.2GHz 5150MHz-5350MHz: 10 dBm (0x50)
        5.2GHz 5350MHz-5470MHz: 10 dBm (0x50)
        5.2GHz 5470MHz-5725MHz: 10 dBm (0x50)
        5.2GHz 5725MHz-5950MHz: 10 dBm (0x50)

 Chain B:
        SAR table index: 1
        2.4GHz:                 13 dBm (0x68)
        5.2GHz 5150MHz-5350MHz: 11 dBm (0x58)
        5.2GHz 5350MHz-5470MHz: 10 dBm (0x50)
        5.2GHz 5470MHz-5725MHz: 10 dBm (0x50)
        5.2GHz 5725MHz-5950MHz: 10 dBm (0x50)


C:\Users\RS214974_DVT1P10209\Desktop\gitit ANT>
```

Figure 28: Generic pSensor Tx Limits

## 8.21 Get Tx Power Delta Limits

Showing the following information:

- GEO table Index.
- Per each band, Tx power delta:
  - Max power
  - Chain A
  - Chain B

Command line options:
ANT –get_tx_power_delta_limits

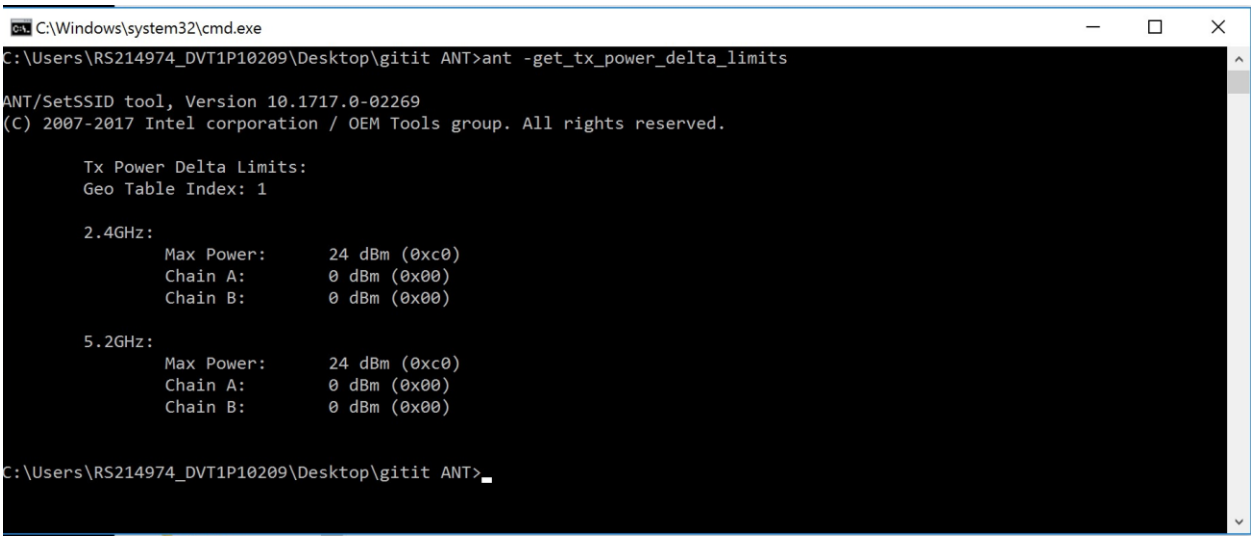



Figure 29: Tx Power Delta Limits

## 8.22 Get BIOS SPLC value

Shows the BIOS SPLC value.

Command line options:

ANT –get_bios_splc

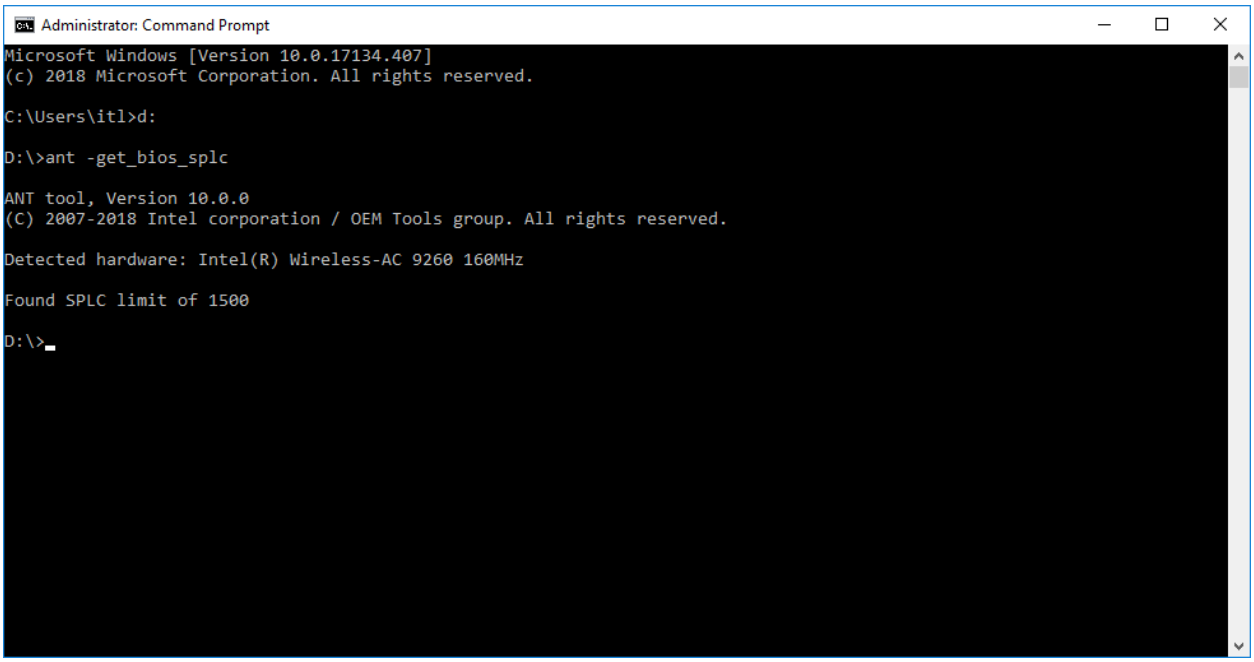



Figure 30: Tx BIOS SPLC value

## 8.23 Get BT BIOS SAR table

Shows the BT SAR values, as read by BT driver.

Command line options:

ANT -get_bt_bios_sar_table

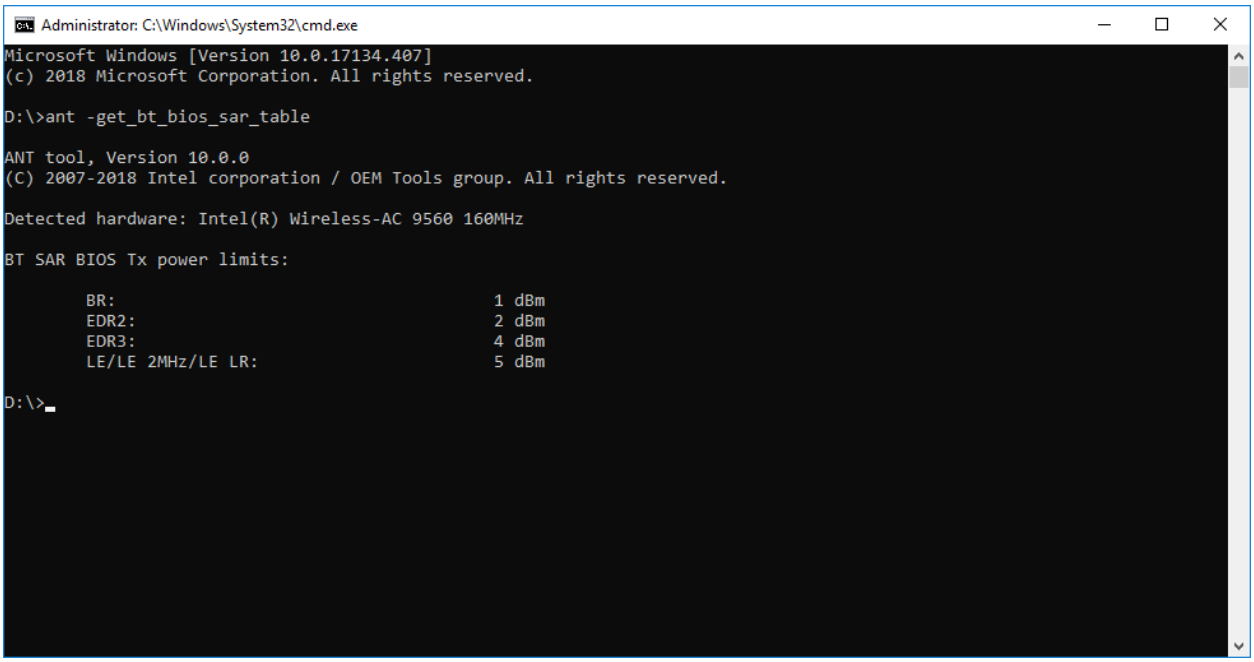



Figure 31: BT BIOS SAR table

- Low energy will display the minimum value between LE, LE 2MHz and LE LR.
- In case BT SAR is disabled in BIOS, OTP power values will be displayed

## 8.24 Get Platform Specific Antenna Gain table

Shows the following information:

- Gain Mode: Enabled \ Disabled.
- Showing the following information for Chain A\B:
  - 2.4GHz
  - 5.2GHz 5150MHz-5350MHz
  - 5.2GHz 5350MHz-5470MHz
  - 5.2GHz 5470MHz-5725MHz
  - 5.2GHz 5725MHz-5950MHz

Command line options:

ANT -get_ant_gain

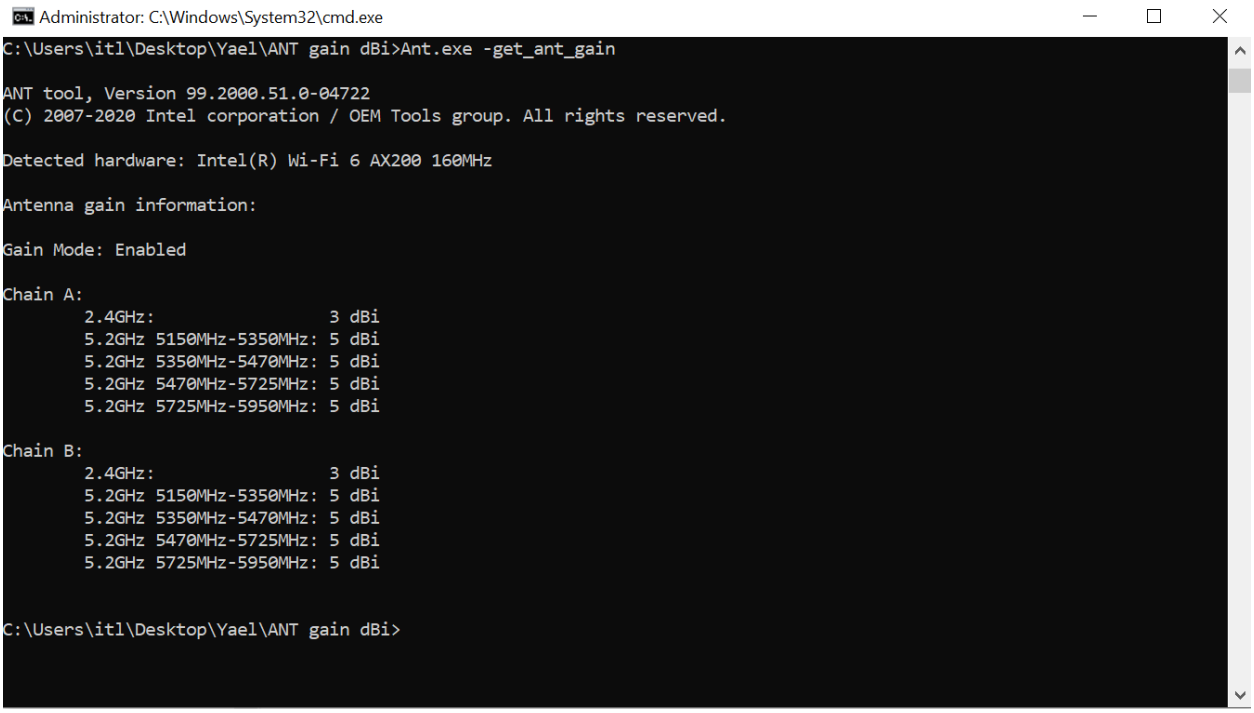



**Figure 32: ANT gain values**

- In case of table information not existing ANT will notify it does not exist.
- In case of table information not valid ANT will notify it is not valid.
- In case of feature being disabled ANT will notify that gain mode is disabled.

## 8.25 Get Time Average SAR (TAS) Information

Shows the TAS information as configured in the platform's BIOS.

- The information presented:
  - TAS Version.
  - Vendor Name.
  - Is Vendor in approved list: Yes / No.
  - Current MCC.
  - UHB for USA Enabled: True / False.
  - TAS MAC0 (For AX411 160MHz the same data will be printed also for MAC1):

| Title | Description | Comments |
|---|---|---|
| TAS Status | Enabled / Disabled | |
| SAR limit value | SAR value in BIOS for this specific location/band | Presented in case Status is Enabled |
| Band | 2.4 GHz / 5 GHz / 6 GHz / Invalid (if not associated to AP) | Presented in case Status is Enabled |
| Max power limit value | The max power of the current location (excluding any SAR limit) | Presented in case Status is Enabled and only when associated to AP |
| Dynamic Status | FW dynamic decision to activate or deactivate TAS (based on some dynamic parameters of the link and mode of operation)<br>Active(5) /Inactive(1/2/3/4)<br>1: MVM Mode<br>2: Trigger Mode<br>3: Black List<br>4: UHB Non US<br>5: Active | Presented in case Status is Enabled and only when associated to AP |
| Near Disconnection | False/True. To indicate WiFi operation in edge of cell (far range). In this case TAS mode might be changed to optimize performance | Presented in case Status is Enabled and only when associated to AP |

  - List of all blocked countries.

Command line options:

- ANT -get_tas_info
- ANT –network <SSID> -get_tas_info

- ANT –network <SSID> -bssid <BSSID> -get_tas_info

```
 TAS Version:                          1
 Vendor Name:                          Intel Corporation
 Is vendor in approved list:           Yes
 Current MCC:                          US
 UHB for USA enabled:                  False

 TAS MAC0:
   Tas Status:                         Enabled
   SAR limit value:                    0 dBm
   Band:                               Invalid
   Max power limit value:              0 dBm
   Dynamic Status:                     Inactive(1)
   Near Disconnection:                 False

 TAS MAC1:
   Tas Status:                         Enabled
   SAR limit value:                    9 dBm
   Band:                                5  GHz
   Max power limit value:              20 dBm
   Dynamic Status:                     Active(5)
   Near Disconnection:                 False

Blocked countries: JP , IL


C:\Users\itl\Desktop\ANT_01038_99.0.77G\ANT>
```

Figure 33: ANT TAS Info

- ANT will notify if the WTAS table does not exist in the BIOS.
- ANT will notify if the TAS values are not valid.
- ANT will notify if the TAS Information feature is disabled.
- In case Association to AP fails, ANT will notify and continue reading from BIOS

## 8.26 Get HW Manufacture Information

Returns who it the manufacture for the specific NIC.
Option is relevant only for HrP & GfP cards.
Possible values:

- <HW ID>
- <HW ID>V2

Command line options:

- ANT -get_hw_manufacture

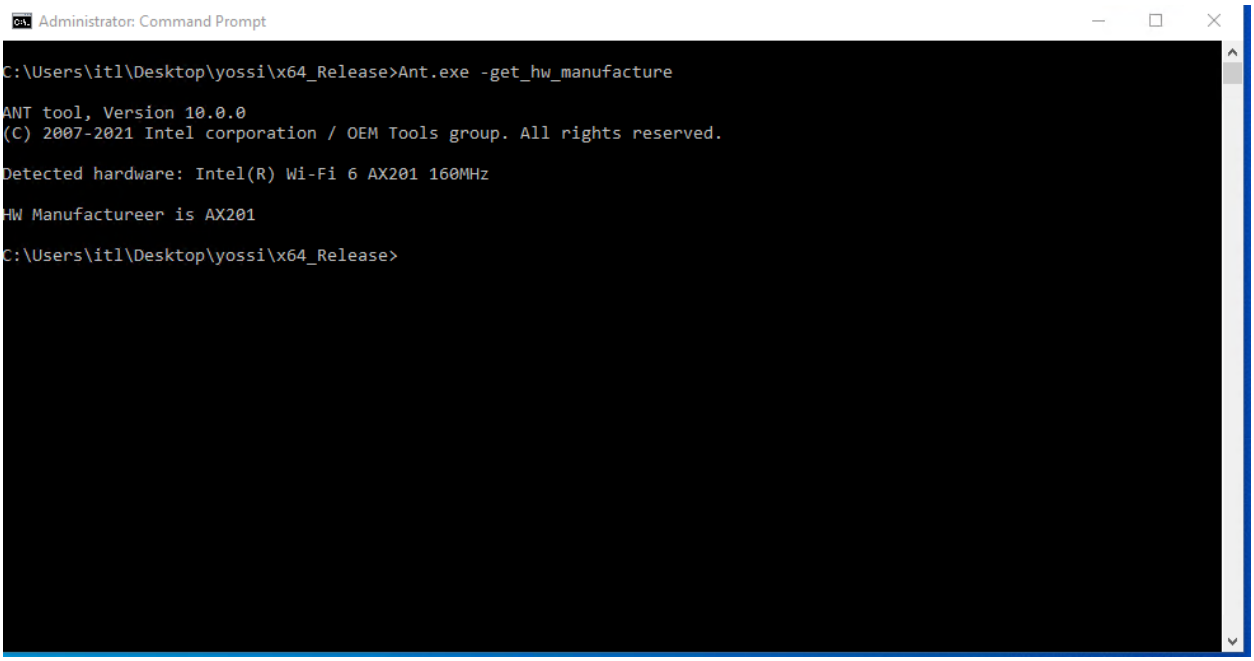



Figure 34: Get HW Manufacture Information

- ANT will notify in case the feature is not supported for the current HW.
- ANT will notify in case the HW manufacture is invalid.

## 8.27 Get Per Country 6e BIOS Settings

Shows 6E country bit status of the BIOS

Command line options:

- ANT -get_per_country_6e_bios_settings

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22000.795]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd C:\Users\Administrator\Desktop\22.160.0.3G\PHWFW06629_22.160.0.3Tools\OEM Tools\ANT

C:\Users\Administrator\Desktop\22.160.0.3G\PHWFW06629_22.160.0.3Tools\OEM Tools\ANT>Ant.exe -get_per_country_6e_bios_settings

ANT tool, Version ANT.00561.22.160.0
(C) 2007-2022 Intel corporation / OEM Tools group. All rights reserved.

Detected hardware: Device Identified by crf_id: 0x0b
SetDeviceFromString: GARFIELD4
Intel(R) Wi-Fi 6E AX411 160MHz

Get board id isnt support by this device
Country             | UHB enabled
----------------------------------------
US                  | Enable
Rest of the world   | Disable
EU                  | Disable
South Korea         | Disable
Brazil              | Disable
Chile               | Disable
Japan               | Disable
Canada              | Disable
```

Figure 35: Get Per Country 6e BIOS Settings

- ANT will notify if command is not supported by current driver.
- ANT will notify if OEM UHB allowed bitmap is not supported by current HW.
- ANT will notify if OEM UHB per country support is disabled in BIOS.

## 8.28 Get Board ID

Shows the SKU Antenna Gain type for AX211.
Possible values:

- Usual – Non High Antenna Gain
- High – High Antenna Gain

Command line option:

- ANT –get_board_id

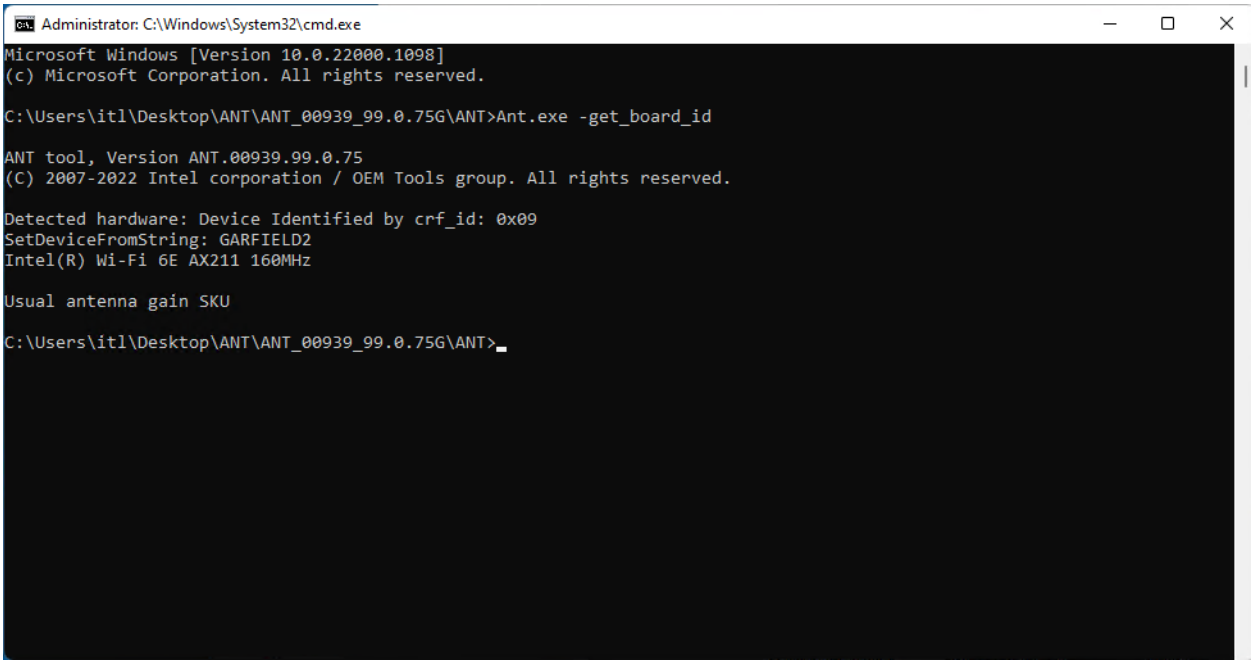



Figure 36: Get Board ID

- ANT will notify in case the feature is not supported for the current HW.
- ANT will notify in case the board ID is not valid.

# Command Line Interface

ANT supports command line interface through a console application.

## 8.29 Command Line Syntax

```
ANT.exe [-option1 <parameter1>] [-option2 <parameter2>] [-option3 <parameter3>] […]
```

For example: ANT –network ssid_name –ant_all

## 8.30 Options and Parameters

For RSSI test it is mandatory to define "-network <ssid>" parameter in order to specifies the AP to associate with.
For setting MCC it is mandatory to define "-SetMcc <mcc>" parameter.

The list below includes all the options and possible values supported by ANT.

| Command line option | Allowed parameter values | Example/comments |
|---|---|---|
| **-? \| /? \| -h \|--help** | | Displays help |
| **-network** | <SSID> | Specifies the AP to associate with |
| **-bssid** | <BSSID> | Specifies the AP MAC address to associate with (Format XX-XX-XX-XX-XX-XX). ANT will look for APs with SSID specified by "-network" and between these will look for an AP with the given BSSID. |
| **-password** | <pass> | Defines the network security key (AP's password). |
| **-band** | <a\|b\|g> | Defines the desired band.<br><br>NOTE: using the "-band" flag affects the scanning bands and may cause the third part applications not see all available stations around. In order to avoid this, use "ant –reset_driver" after finishing the tests. |
| **-antd** | | Enables antenna diversity for all antennas |
| **-ant1** | | Performs RSSI test on ant 1/A |
| **-ant2** | | Performs RSSI test on ant 1/B |
| **-ant_all** | | Performs RSSI test on all available antennas |
| **-enable_11n** | | enable 11n mode |

| | | |
|---|---|---|
| **-disable_11n** | | disable 11n mode |
| **-i** | <1…100><br>Default value is 1 | Defines the number of RSSI tests to run |
| **-th** | <10…200><br>Default value is 45 | Defines the RSSI threshold value to use in the test.<br><br>NOTE: Use a positive value eg.43, program will multiply by -1 result eg.-43) |
| **-th_high** | <0…200><br>Default value is 0 | Defines the maximum RSSI threshold value to use in the test. |
| **-deltaab** | <0…200> | Defines delta threshold for delta test between A and B antennas |
| **-rssi_loops** | <1…100><br>Default value is 10 | Defines the number of RSSI samples per test. |
| **-stay_assoc** | | Cancels the disassociation process at the end of the ANT process. |
| **-falloff** | Default value is 3 | Sets fall off parameter of waiting for stable RSSI. This is the weight ratio of accumulated RSSI compared to weight of current RSSI when waiting for a stable RSSI. |
| **-stable_rssi_threshold** | Default value is 1 | Sets threshold for the diff between current RSSI and average RSSI when waiting for stable RSSI. |
| **-delay** | Default value is 1 msec | Sets delay, in msec between changing band/diversity/11n mode parameters and starting association. |
| **-enable_driver** | | Enables the Intel® Wi-Fi driver |
| **-disable_driver** | | Disables the Intel® Wi-Fi driver |
| **-reset_driver** | | Resets the Intel® Wi-Fi driver |
| **-enable_bt_driver** | | Enables the Bluetooth driver |
| **-disable_bt_driver** | | Disables the Bluetooth driver |
| **-get_rf_kill** | | Returns the status of HW RF-Kill |
| **-xml** | <o\|a> | Extracts test output to xml file (supports only RSSI tests).<br>A – Appends to an existing XML.<br>O – Overwrite an existing XML (or creating new XML if not existing). |
| **-OutputXMLFile** | <filename><br>Default file name "AntennaAppSummary.xml" | Sets output XML file to be used by "-xml". |
| **-d \| -debug** | <append\|overwrite> | Output logs to 'antToolLog.log'. Overwrite will delete all file content |
| **-log** | <filename> | Extract significant values to file. |
| **-BT** | | Measures BT RSSI from air |
| **-SetMcc** | <mcc> | Sets the requested mcc for 2 hours |

| | | |
|---|---|---|
| | | Example: ANT –SetMcc US |
| **-QueryMcc** | | Returns the current MCC, MCC source and If the source is from MCC API |
| **-wev_off** | | Disables MCC events. |
| **-msg_box_off** | | Disables MCC popups messages. |
| **-aggressive** | | Sends association requests once every 2 seconds until association succeeds or time out expires. |
| **-get_hw_ver** | | Gets HW version.<br>Supported only for WP device! |
| **-compare_hw_ver** | <HW_VERSION_COMPARING_WITH> | Compares HW version.<br>Supported only for WP device!<br>Example: ANT –compare_hw_ver 1.1.40 |
| **-get_deviceid** | | Gets device ID |
| **-get_subsystemid** | | Gets subsystem ID |
| **-get_nvmimage** | | Gets NVM image version |
| **-get_oemversion** | | Gets OTP OEM version |
| **-get_device_data** | | Gets device data |
| **-get_pcirevid** | | Gets PCI revision ID |
| **-get_psensor_tx_limits** | | Gets Lenovo pSensor Tx Limits.<br>Returns 2.4 and 5.2 GHz Tx Power Limit in case of WMI or HID sensor.<br>Returns TX limit values for Chain A and B:<br>• 2.4GHz<br>• 5.2GHz 5150MHz-5350MHz<br>• 5.2GHz 5350MHz-5470MHz<br>• 5.2GHz 5470MHz-5725MHz<br>• 5. 5.2GHz 5725MHz-5950MHz<br>In case of EXTENDED HID sensor. |
| **-get_psensor_full_tx_limits** | | Gets Generic pSensor Tx Limits.<br>Returns the Index of chain A/B SAR table in use and TX limit values for Chain A and B:<br>• 2.4GHz<br>• 5.2GHz 5150MHz-5350MHz<br>• 5.2GHz 5350MHz-5470MHz<br>• 5.2GHz 5470MHz-5725MHz<br>• 5. 5.2GHz 5725MHz-5950MHz |
| **-get_bios_sar_table** | | Gets BIOS SAR Table. |

| | | |
|---|---|---|
| | | Returns table for chain A and B:<br>• 2.4GHz<br>• 5.2GHz 5150MHz-5350MHz<br>• 5.2GHz 5350MHz-5470MHz<br>• 5.2GHz 5470MHz-5725MHz<br>• 5. 5.2GHz 5725MHz-5950MHz |
| **-get_tx_power_delta_limits** | | Gets Tx Power Delta Limits.<br>Returns the index of the table in use and per each band, Tx power delta:<br>• Max power<br>• Chain A<br>• Chain B |
| **-old_ant_map** | | ANT was updated to detect HW type and note the relevant antenna port labeling. This option will use the legacy report data and format |
| **-wifi_on** | | Enables Wi-Fi radio (Set radio state to true) |
| **-ant_isolation** | | Shows the physical isolation between two antennas when Wi-Fi is associated to AP (2.4GHz only) and BT is active. Antenna isolation value is in dB. |
| **-scan** | | Single channel scan mode. |
| **-channel** | < active channel number > | The scan is active only, thus only active channels are supported.<br>Acceptable channels: 1-11, 36, 40, 44, 48, 149, 153, 157, 161 and 165 |
| **-interval** | <time in ms> | Time interval of every iteration in millisecond (ΔT –time between scans). Allowed values: 0, 200, 400, 600, 800 and 1000. Default is `200ms`. |

# 9. Error codes and return values

| Error code # | Error description |
|---|---|
| 0 | Test passed |
| 1 | Test failed |
| 2 | Association failed |
| 3 | No appropriate installed driver was found |
| 4 | HW RF-Kill is enabled |
| 5 | HW RF-Kill is disabled |
| 6 | Bluetooth not supported |
| 7 | Hardware version not supported |
| 8 | MCC invalid time |
| 9 | MCC FW denial |
| 10 | MCC already set |
| 11 | Failed to enable Bluetooth driver |
| 12 | Failed to disable Bluetooth driver |
| 13 | No relevant BIOS table exist |
| 14 | FW was not configured with SAR BIOS settings |
| 15 | No sensor exist |
| 16 | Sensor type not supported |
| 256 | Invalid command line |
| 257 | Hardware initialization error |
| 258 | General error |
| | |

# 10. Appendix

## 10.1 Acronyms

| Acronym | Description |
|---|---|
| AP | Access Point |
| RSSI | Received Signal Strength Indicator |
| CLI | Command Line Interface |
| MCC | Mobile Country Code |
| | |
| | |
| | |
| | |
| | |
| | |

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